StoryCast

Project Documentation

An accessible audio and video storytelling microsite
Semantic HTML5 · Sass · CSS Grid and Flexbox · vanilla JavaScript

Prepared by Chibueze Nnamdi Onugha
African Leadership University · Software Engineering
October 2026

1. Overview

StoryCast is a static site with no framework and no runtime dependencies. Sass compiles to a single stylesheet (css/main.css) and one small script (js/main.js) adds optional enhancements. This document describes how the project is organised, how the design tokens and components work, how each accessibility requirement was met, how it was tested, and how to run and extend
it.

2. Project structure

storycast/
├── index.html · about.html
├── story/ three story detail pages
├── sass/ main.scss + partials (tokens, base, layout, components)
├── css/main.css compiled output
├── js/main.js progressive enhancement
├── assets/ audio/ video/ captions/ transcripts/ images/
├── docs/ PDFs, sitemap, wireframes, screenshots
└── README.md Detailed information and instructions about the site

3. Sass architecture

Styles are written with Sass modules (@use). main.scss only imports partials. The three token partials hold the design decisions; _base.scss turns them into CSS custom properties on :root so they are available at run time and easy to theme.

Partial	Responsibility
_colors.scss	Palette map and the clr() helper that returns var(--color-*).
_typography.scss	Font stacks, fluid type scale, line heights, reading measure
_spacing.scss	Spacing scale, breakpoints (em), container width, gutter, radius, touch-target size.
_mixins.scss	up() breakpoint mixin and visually-hidden mixin.
_base.scss	Custom properties, reset, typography, links, focus styles, reduced-motion, forced-colors and print rules
_layout.scss	Page frame, header and nav (Flexbox), hero, card grid, story layout, feature grid, footer (Grid).
_components.scss	Skip link, buttons, badge, media card (container query), player, transcript (container query), callout, table.
_utilities.scss	visually-hidden, stack, cluster, prose, text-soft

4. Key components

4.1 Media card (container query)

The card sets the container: media-card / inline-size, so its layout depends on the space it is given, not the viewport. The same markup is used full-width on Home, in a three-column grid, and in the 20rem related-stories sidebar.

4.2 Player and captions

Players use native controls inside a <figure> with a <figcaption>. Captions come from a WebVTT file attached with <track kind="captions" default>. Browsers do not display captions for <audio>, so audio stories mirror the current caption in a visible line under the player (added by JavaScript) and the full transcript is always on the page.

4.3 Transcript

The transcript is a labelled, keyboard-focusable scroll region. Each cue carries data-start and data-end in seconds. With JavaScript, each timestamp becomes a button that seeks the media and the line being spoken gets aria-current="true" plus a visual bar. Without JavaScript the transcript is still complete and readable. A second container query moves timestamps into their own column once the transcript is 30rem wide.

4.4 JavaScript (progressive enhancement)

• Upgrade <time> stamps to buttons that seek the media.
• Highlight the current line and mirror it as a live caption for audio.
• Playback speed (0.75× to 1.5×) and an optional follow-along scroll that respects reduced motion.
• About 90 lines, no libraries. If the script fails, nothing essential is lost.

5. Run it locally

```
git clone https://github.com/Nnamdi004/storycast.git
cd storycast
npm install       # installs Sass
npm run build     # sass/main.scss -> css/main.css
npm run serve     # http://localhost:8000
```

Use a static server that supports HTTP range requests (the serve script does). Python's built-in http.server does not, which stops transcript timestamps from seeking. npm run watch recompiles Sass on every save

6. Adding or replacing a story

• Add media to assets/audio or assets/video, captions to assets/captions (WebVTT) and a text transcript to assets/transcripts.
• Copy an existing story page; update the title, media paths, transcript cues (start/end seconds) and related stories.
• Add a card for it on index.html.

7. Known limitations

• Sample narration is synthetic and the video visuals are abstract placeholders. Replace them with real media.
• Captions are English only. Add more <track> elements for other languages.
• The site is light-theme only; a dark theme and a high-contrast option are natural next steps.
• Figma wireframe and mockup links are not included in this repository yet. SVG wireframes and built-page screenshots are provided for import and reference.